Understand Powers and Exponents

Be Curious

Which Doesn't Belong?

7×7

4×4

6×8

10×10

Sample explanation

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Math is... Mindset

What do you want your classmates to know about your math story?

Learn

At Week 1, Dean had 10 pennies Each week after, Dean increased the number of pennies by 10 times the previous week.

During which week will Dean have 1,000,000 pennies?

You can organize the information in a table to help determine the solution.

Math is... Patterns

What patterns do you notice in the table?

Week	Multiplication Expression	Number of Pennies Added each Week
1	10	10
2	10×10	100
3	$10 \times 10 \times 10$	1,000
4	$10 \times 10 \times 10 \times 10$	10,000
5	$10 \times 10 \times 10 \times 10 \times 10$	100,000
6	$10 \times 10 \times 10 \times 10 \times 10 \times 10$	1,000,000

A **power of 10** is the product of 10 multiplied by itself a number of times.

Dean will have 1,000,000 pennies in Week 6.

You can write a power of 10 as a multplication expression with factors of 10.

You can also write a power of 10 in **exponential form** using a **base** and an **exponent**.

$\underbrace{10 \times 10}_{\text{2 factors}} = 10^{2}$ ← exponent

↑ base

Work Together

Write 10^{8} as a multiplication expression. Then, find the product.

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On My Own

MATH REPLAY | GO ONLINE

Name _____

Write the exponential form as a multiplication expression.

1. 10^4 $10 \times 10 \times 10 \times 10$
2. 10^2
3. 10^3
4. 10^6

Write the exponential form.

5. $10 \times 10 \times 10 =$ _____
6. $10 \times 10 \times 10 \times 10 \times 10 =$ _____
7. $10 \times 10 \times 10 \times 10 =$ _____
8. $10 \times 10 =$ _____

Write the exponential form of each power of 10.

9. $10 =$ _____
10. $1{,}000 =$ _____
11. $100 =$ _____
12. $10{,}000 =$ _____

13. Rachel finds the value of 10^5 as shown. Do you agree with her solution? Tell why.

$10^5 = 10 \times 5 = 50$

14. **STEM Connection** Grace reviewed 10^6 lines of a computer program. How many lines did she review? Write the product.



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15. Trevor's personal walking goal is shown on his activity tracker.



A. How can you help Trevor write this goal using an exponent?

B. How can you help Trevor write this goal as a product of 10s?

16. Jenny's father gives her $10. Her grandfather offers to give her ten times the value her father gave her. Her grandmother offers her a choice of either ten times the value her grandfather offered or $500. Which of her grandmother's offers should Jenny choose? Explain.

$1,000

17. Extend Your Thinking Consider the inequality shown.

$$10 \times 10 \times 10 \times 10 < b < 10^6$$

What is the value of b? Explain.

Reflect

What patterns did you notice when writing different forms of powers of 10?

Math is... Mindset

How have you told your classmates about your math story?

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